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Chapter 1

Schwarzian derivatives and cylinder maps

In [1], the authors constructed a family of skew product diffeomorphisms whose dynamic properties vary according to the Schwarzian derivative of the maps on the fibers. Depending on the sign of this derivative, the family either has intermingled basins, a physical measure whose basin has full Lebesgue measure, or historical behavior almost everywhere. During the first year, we sought to understand this family of examples well and answer some questions posed in the article.

1.1 Cross ratios and the Schwarzian derivative

Suppose we start with a field $\mathbb{K}$ and, not being happy with the fact that we cannot divide by zero, decide to add a point at infinity; this gives us the *projective line* $P^1(\mathbb{K}) = \mathbb{K} \sqcup \{\infty\}$, whose arithmetic operations are the same as those of $\mathbb{K}$ except for the additional rules $\frac{1}{0} = \infty$, $\frac{1}{\infty} = 0$, $x \cdot \infty = \infty$ if $x \neq 0$, and $x + \infty = \infty$ if $x \neq \infty$. It's not possible to define such operations everywhere without introducing inconsistent rules, so the operations are partial — we cannot multiply $\infty \cdot 0$ or add $\infty + \infty$.

Our new space $P^1(\mathbb{K})$ has an interesting property: given $z \in P^1(\mathbb{K})$, we have three types of operations that we can apply to z: translations $z \mapsto z + x$, where $x \in \mathbb{K}$, homotheties $z \mapsto xz$ for $x \in \mathbb{K} \setminus \{0\}$, and the inverse $z \mapsto 1/z$, which was not defined before if $z = 0$, but now makes sense for every z. These three families of actions, which are invertible, generate the *group of homographies* with coefficients in $\mathbb{K}$.

A second way to construct $P^1(\mathbb{K})$, which makes the role of the group of homographies as a symmetry group of this space even clearer, is to think of the space of "formal fractions", that is, pairs of numbers (x, y), where $x, y \in \mathbb{K}$ are not both zero, where we identify "equivalent fractions", in the sense that $(\lambda x, \lambda y) \sim (x, y)$ for every $\lambda \in \mathbb{K} \setminus \{0\}$. The quotient by this equivalence relation gives us $P^1(\mathbb{K})$ as the space of lines passing through the origin of the vector space $\mathbb{K}^2$, and the operations are like fraction operations: $(x, y) + (z, w) = (wx + yz, yw)$, $(x, y) \cdot (z, w) = (xz, yw)$, and $(x, y)^{-1} = (y, x)$. Despite being seemingly more complicated, the interesting fact about this construction is that it emphasizes how the symmetry group of the space $\mathbb{K}^2$, the linear group $\mathrm{GL}_2(\mathbb{K})$, naturally induces a symmetry group on the quotient, the projective group $\mathrm{PGL}_2(\mathbb{K})$, which is precisely the group of homographies.

This group is a quotient of $\mathrm{GL}_2(\mathbb{K})$ by a subgroup of dimension 1, so it has dimension 3. Furthermore, it is 3-transitive: given two sets of three distinct points in $P^1(\mathbb{K})$, there exists an element $g \in \mathrm{PGL}_2(\mathbb{K})$ that takes the first set to the second. But not only that, $P^1(\mathbb{K})$ has three special points: 0, 1, and ∞. Thus, given $x_1, x_2, x_3 \in P^1(\mathbb{K})$ distinct, we can ask which

$g \in \mathrm{PGL}_2(\mathbb{K})$ such that $g(x_1) = \infty$, $g(x_2) = 0$, and $g(x_3) = 1$. With some calculations, we arrive at

$$g(z) = \frac{(x_2 - z)(x_3 - x_1)}{(x_1 - z)(x_3 - x_2)}$$

The *cross ratio* of four points x_0, x_1, x_2, x_3 is the value of $g(x_0)$ for the element g of the homography group that sends $x_1 \mapsto \infty$, $x_2 \mapsto 0$, and $x_3 \mapsto 1$.

Definition 1 (Cross ratio)**.**

$$\rho(x_0, x_1, x_2, x_3) = \frac{(x_2 - x_0)(x_3 - x_1)}{(x_1 - x_0)(x_3 - x_2)}$$

Definition 2. Let $f\colon I \to I'$ be a C^3 diffeomorphism between two intervals of the real line. The *Schwarzian derivative* of f is the function

$$\mathcal{S}f = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2.$$

We shall write $\mathcal{S}f \prec 0$ if there exists a dense open set $U \subseteq I$ such that $\mathcal{S}f(y) < 0$ for every $y \in U$. Similarly, we write $\mathcal{S}f \succ 0$ if there exists $U \subseteq I$ open and dense such that $\mathcal{S}f(y) > 0$ for every $y \in U$.

Proposition 1. *The Schwarzian derivative has the following properties:*

1. *If $\mathcal{S}f$ and $\mathcal{S}g$ have the same sign, then $\mathcal{S}(g \circ f)$ has the same sign.*
2. *If $\mathcal{S}f \prec 0$, then $\mathcal{S}f^{-1} \succ 0$.*
3. *$\mathcal{S}f \prec 0$ if, and only if, $\varphi(y) = 1/\sqrt{|f'(y)|}$ is strictly convex.*
4. *$\mathcal{S}f \equiv 0$ if, and only if, f is a fractional linear transformation.*

Lemma 2. *If $f\colon I \to I'$ is C^3, then $\mathcal{S}f \prec 0$ if, and only if, f increases cross ratios, in the sense that $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$ for all $y_1 < y_2 < y_3 < y_4$.*

Lemma 3. *If $f\colon I \to I$ preserves orientation and $\mathcal{S}f \prec 0$, then $f'(0)f'(1) < 1$.*

Note that by proposition 1, analogous results to the last two lemmas hold if $\mathcal{S}f \succ 0$. For example, if $\mathcal{S}f \succ 0$ then f decreases cross ratios and $f'(0)f'(1) > 1$.

1.2 Skew products

We call a map $F\colon X \times Y \to X \times Y$ a *skew product* if it is of the form $F(x, y) = (g(x), f_x(y))$ for some $g\colon X \to X$ and a family of functions $\{f_x\}_{x \in X}$, $f_x\colon Y \to Y$. We are interested in the case where X is a compact metric space with the Borel σ-algebra, $g\colon X \to X$ is a continuous map preserving an ergodic probability measure $\mu \in \mathcal{M}(X)$, $Y = I$ is the interval $[0, 1]$, and $f_x\colon I \to I$ are C^3 diffeomorphisms preserving orientation. We will see how, depending on the Schwarzian derivative of the maps f_x, we observe different behaviors of the dynamics with respect to the invariant measures $\nu_0 = \mu \otimes \delta_0$ and $\nu_1 = \mu \otimes \delta_1$. Note that for $i \in \{0, 1\}$, $\operatorname{supp} \nu_i = \mathcal{A}_i$, where $\mathcal{A}_i = X \times \{i\}$ are the boundaries of the "cylinder" $X \times I$. Let $\mathcal{B}_i = \mathcal{B}(\nu_i)$ be the basins of the respective measures.

To simplify notation, let's denote by $f_x^n(y)$ the second coordinate of $F^n(x, y)$, that is, $f_x^n = f_{g^{n-1}(x)} \circ \cdots \circ f_x$.

Lemma 4. *If $(x, y_1) \in \mathcal{B}_0$, then $(x, y_2) \in \mathcal{B}_0$ for all $y_2 \leq y_1$. Similarly, if $(x, y_1) \in \mathcal{B}_1$, then $(x, y_2) \in \mathcal{B}_0$ for all $y_2 \geq y_1$.*

Proof. We will prove the case $i = 0$, as the situation is symmetric. Let $(x, y_1) \in \mathcal{B}_0$ and $y_2 \leq y_1$. If $\varphi \in C(X \times I)$, and $\varepsilon > 0$, by the uniform continuity of φ, there exists $\delta > 0$ such that $y < \delta$ implies $|\varphi(x, 0) - \varphi(x, y)| < \varepsilon$ for all $x \in X$. Since $U = X \times [0, \delta)$ is a continuity set for ν_0 and $e_n(x, y_2) \to \nu_0$, by Portmanteau

$$\lim_{n\to\infty} e_n(x, y_2)(U) = \nu_0(U) = 1.$$

Define $\tau_n = e_n(x, y_2)(U) = \frac{1}{n}\#\{0 \leq i < n \mid f_x^n(y_2) < \delta\}$. Note that if $f_x^n(y_2) < \delta$, then $f_x^n(y_1) < \delta$ since the diffeomorphisms preserve orientation. Thus,

$$\begin{aligned}|e_n(x, y_1)(\varphi) - e_n(x, y_2)(\varphi)| &\leq \frac{1}{n}\sum_{i=0}^{n-1}\left|\varphi(g^n(x), f_x^n(y_1)) - \varphi(g^n(x), f_x^n(y_2))\right| \\ &\leq \varepsilon\tau_n + 2\|\varphi\|_\infty(1 - \tau_n).\end{aligned}$$

Since $\tau_n \to 1$, there exists N such that $n \geq N$ implies that the right side is less than 2ε, hence both integrals on the left converge to $\nu_0(\varphi)$. □

Transversal Lyapunov exponents

To each point (x, y) of either boundary $\mathcal{A}_i$, for $i \in \{0, 1\}$, we can associate a number that expresses the average expansion rate of tangent vectors along the direction normal to the skew product boundary. This is a specific case of the general definition of Lyapunov exponents, whose existence is given by the Oseledets theorem.

Definition 3 (Transversal Lyapunov exponent)**.**

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \lim_{n\to\infty} \frac{1}{n} \log|\partial_y F^n(x, i)|.$$

By the product rule, and noting that $\{0, 1\}$ are fixed points of the fiber diffeomorphisms, the right side of the equation above can be rewritten as

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} \log\left|\partial_y F(g^k(x), i)\right| = \lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} \log f'_{g^k(x)}(i).$$

Thus, by the ergodicity of μ for g and the Birkhoff ergodic theorem, for μ-almost every $x \in X$ it holds that

$$\mathrm{Lyap}_{\mathcal{A}_i}(x) = \int \log f'_x(i)\, d\mu(x), \tag{1.1}$$

and we shall denote this common value by $\mathrm{Lyap}(\mathcal{A}_i)$.

The version of the statement below is slightly more general than the corresponding one in [1, lema A.1], that asks for F to be C^2. In our context, $F(x, y) = (g(x), f_x(y))$ where g is an ergodic map for μ and $f_x \in C^1[0, 1]$ for all x. In particular, if $(x, y) \mapsto f'_x(y)$ is continuous, the following equicontinuity hypothesis is satisfied.

Lemma 5. *If the family $\{f'_x\}_{x\in X}$ is equicontinuous at $i \in \{0, 1\}$ and $\mathrm{Lyap}(\mathcal{A}_i) < 0$, then the basin of $\mu \otimes \delta_i$ has positive $\mu \otimes \mathrm{Leb}$ measure. Conversely, if $\mathrm{Lyap}(\mathcal{A}_i) > 0$, then the corresponding basin has null measure.*

Proof. We will assume $i = 0$. Let $\varepsilon > 0$ be such that

$$\int \log(f_x'(0) + \varepsilon)\, d\mu(x) < 0.$$

Let $a(x) = \log(f_x'(0) + \varepsilon)$. By the Birkhoff ergodic theorem, for μ-almost every $x \in X$

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} a(g^k(x)) = \int a(x)\, d\mu(x) < 0,$$

which implies $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$. This means that the function

$$A(x) = \max_{n\geq 1} \sum_{k=0}^{n-1} a(g^k(x)) \geq 0$$

is well-defined in almost every x.

By equicontinuity, there is $\delta > 0$ such that $0 \leq y \leq \delta$ implies $f_x'(x) \leq f_x'(0) + \varepsilon$ for all $x \in X$. Thus, if $y < \delta e^{-A(x)}$, we claim that

$$f_x^n(y) \leq \exp\left(\sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta \leq \delta. \tag{1.2}$$

is true by induction. Indeed, the $n = 0$ case follows from the hypothesis on y, and if (1.2) is true for n, then by the mean value theorem,

$$\begin{aligned} f_x^{n+1}(y) &\leq \left(\max f'_{g^n(x)}([0, f_x^n(y)])\right) f_x^n(y) \\ &\leq (f'_{g^n(x)}(0) + \varepsilon) f_x^n(y) \\ &\leq \exp\left(a(g^n(x)) + \sum_{k=0}^{n-1} a(g^k(x)) - A(x)\right)\delta. \end{aligned}$$

Since $\lim_{n\to\infty} \sum_{k=0}^{n-1} a(g^k(x)) = -\infty$, we have $f_x^{n+1}(y) \to 0$. Thus, if $x \in \mathcal{B}(\mu)$ (which has full measure), then $e_n(x, y) \Rightarrow \nu_0$ whenever $y \leq \delta e^{-A(x)}$.

WRITE THE CONVERSE □

Corollary 6. *If $(x, y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x \prec 0$ for μ-almost every $x \in X$, then* $\mathrm{Lyap}(\mathcal{A}_0) + \mathrm{Lyap}(\mathcal{A}_1) < 0$. *In particular, at least one of the basins has positive measure.*

Proof. If suffices to integrate the logarithm of the inequality $f_x'(0)f_x'(1) < 1$ from lemma 3, and then apply lemma 5. □

Intermingled Basins

In this section only, we fix $X = S^1$, $\mu = \mathrm{Leb}$, and $g\colon S^1 \to S^1$ the map $g(x) = kx$. Furthermore, we require the following hypothesis to hold.

Hypothesis 1 (Elevators)**.** There exist fixed points x^- and x^+ in S^1 such that for every $y \in (0, 1)$, $f_x(y) < y$ for x in a neighborhood of x^- and $f_x(y) > y$ for x in a neighborhood of x^+.

Definition 4. We say that two measurable subsets A, B are *intermingled* with respect to a measure ν if for every open set U, $\nu(A \cap U)$ and $\nu(B \cap U)$ are strictly positive.

Theorem 7. *Suppose hypothesis 1 holds and both basins have positive measure. Then the basins $\mathcal{B}_0$ and $\mathcal{B}_1$ are intermingled for the Lebesgue measure on the cylinder.*

Proof. By definition, the basins are strictly invariant sets. If we define the measures $\mu_i(A) = \mathrm{Leb}(A \cap \mathcal{B}_i)$, then the basins are intermingled if, and only if, $\operatorname{supp}\mu_i = X \times I$ for both $i = 0, 1$. We will show this is the case for $\operatorname{supp}\mu_0$, as the proof for $i = 1$ is analogous.

Claim 1: The support $\operatorname{supp}\mu_0$ is such that $F^{-1}(\operatorname{supp}\mu_0) \subseteq \operatorname{supp}\mu_0$.

Indeed, if $p \in F^{-1}(\operatorname{supp}\mu_0)$ and U is a neighborhood of p, then since F is an open map, $F(U)$ is a neighborhood of $F(p) \in \operatorname{supp}\mu_0$, so $\mu_0(F(U)) > 0$. Since F is Lipschitz, if $\mu_0(U)$ were 0, then $\mu_0(F(U)) = \mathrm{Leb}(F(U) \cap \mathcal{B}_0) = \mathrm{Leb}(F(U \cap \mathcal{B}_0)) = 0$, which is not the case, so $\mu_0(U) > 0$.

Claim 2: There exists a point $(x_0, y_0) \in \operatorname{supp}\mu_0 \setminus X \times \{0\}$.

Indeed, $\mu_0(\operatorname{supp}\mu_0) \geq \mathrm{Leb}(\mathcal{B}_0) > 0$, so $\mu_0(\operatorname{supp}\mu_0 \setminus X \times \{0\}) > 0$ and the set cannot be empty.

Claim 3: The endpoint $\mathcal{A}_1$ of the cylinder is contained in $\operatorname{supp}\mu_0$.

Inductively, we can construct a backward orbit $\cdots \mapsto x_{-2} \mapsto x_{-1} \mapsto x_0$ of g such that $x_{-n-1} \in g^{-1}(\{x_{-n}\})$ is the closest preimage of x_-. Since x_- is a repelling fixed point, the construction implies that $x_{-n} \to x_-$, so by hypothesis 1, for every $y \in (0, 1)$ there exists n_0 such that $n \geq n_0$ implies $f_{x_{-n}}^{-1}(y_0) > y$. In particular,

$$y_{-n} = f_{x_{-n}}^{-1} \circ \cdots \circ f_{x_{-1}}^{-1}(y_0)$$

is such that $y_{-n} \to 1$. Since $\operatorname{supp}\mu_0$ is closed and $(x_{-n}, y_{-n}) \in F^{-n}(\operatorname{supp}\mu_0)$, we conclude that $(x_-, 1) \in \operatorname{supp}\mu_0$. Now, note that $E = \bigcup_{n=0}^{\infty} g^{-1}(\{x_-\})$ is dense in X and 1 is a fixed point of the f_x. Thus, $E \times \{1\} \subseteq \operatorname{supp}\mu_0$ and consequently $\mathcal{A}_1 \subseteq \operatorname{supp}\mu_0$.

Claim 4: Given $x \in X$, $\operatorname{supp}\mu_0 \cap \{x\} \times I$ is an interval $\{x\} \times [0, a_x]$.

By lemma 4, $\mathcal{B}_0$ is of the above form. To show that the support also has this property, we can take $y < 1$ such that $(x, y) \in \operatorname{supp}\mu_0$ and show that if $z < y$ then $(x, z) \in \operatorname{supp}\mu_0$.

Let $U = B_\varepsilon(x, z)$ be a ball centered at (x, z) of radius smaller than $1 - y$, and $V = B_\varepsilon(x, y)$. Note that $V \cap \mathcal{B}_0 \subseteq (U \cap \mathcal{B}_0) + (0, y - z)$, and since $(x, y) \in \operatorname{supp}\mu_0$, $\mathrm{Leb}(U \cap \mathcal{B}_0) \geq \mathrm{Leb}(V \cap \mathcal{B}_0) > 0$. This shows the claim.

As $\mathcal{A}_1$ is contained in the support, the last claim implies $\operatorname{supp}\mu_0 = X \times I$. □

Example 1.2.1. Define $q_a(y) = y + ay(1 - y)$ and $p(x) = \varepsilon\cos(2\pi x)$ for some $0 < \varepsilon < 1$. If $|a| < 1$, then q_a is a diffeomorphism of the interval I. If $F(x) = (3x, q_{p(x)}(y))$, then $x^+ = 0$ and $x^- = 1/2$ are fixed points of the map $g(x) = 3x$ (where we identify S^1 with $[0, 1]$) that satisfy hypothesis 1, because $q_a(y) > y$ if $a > 0$ and $q_a(y) < y$ if $a < 0$ for all $y \in (0, 1)$. Moreover, since q_a is a degree two polynomial, it has negative Schwarzian whenever $a \neq 0$, and from the symmetry of p we see both basins of $F(x, y) = (kx, q_{p(x)}(y))$ must have positive measure, because at least one of them has. So F displays intermingled basins.

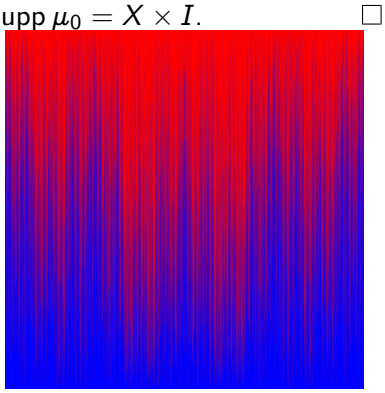

Figure 1.1: Basins of example 1.2.1.

Negative Schwarzian

The statement of the theorem below corresponds to [1, Theorem 3.2] without the hypothesis that both basins have positive measure. The question of whether this hypothesis could be removed was left open in the article, and the following proof shows that it is indeed unnecessary.

Theorem 8. *If $(x, y) \mapsto \mathcal{S}f_x(y)$ is continuous and $\mathcal{S}f_x \prec 0$ almost everywhere in $x \in X$, then there exists a measurable function $\sigma\colon X \to I$ defined almost everywhere such that*

$$(x, y) \in \mathcal{B}_0 \text{ whenever } y < \sigma(x) \quad \text{and} \quad (x, y) \in \mathcal{B}_1 \text{ whenever } y > \sigma(x).$$

Consequently, the union $\mathcal{B}_0 \cup \mathcal{B}_1$ has total $\mu \otimes \mathrm{Leb}$ measure.

Proof. We start by defining functions $\xi_0, \xi_1\colon X \to I$ that separate the basins in the following sense:

$$\xi_0(x) = \sup\{y \in I : y \in \mathcal{B}_0\}, \qquad \xi_1(x) = \inf\{y \in I : y \in \mathcal{B}_1\}.$$

By lemma 4, if $y < \xi_0(x)$ or $y > \xi_1(x)$, the empirical measures $e_n(x, y)$ converge to $\nu_0 = \mu \otimes \delta_0$ or to $\nu_1 = \mu \otimes \delta_1$, and if $\xi_0(x) < y < \xi_1(x)$, they do not converge to either ν_0 or ν_1.

It is sufficient to show that $A = \{x \in X : \xi_0(x) \neq \xi_1(x)\}$ has measure zero, since in this case we can define σ as the common value of the two functions. Suppose by contradiction that this set does not have measure zero. Since the f_x are bijective, A is an invariant set under g, so the ergodicity of μ implies that A has full measure.

Since we are under the hypotheses of corollary 6, we can assume without loss of generality that $\mathrm{Leb}(\mathcal{A}_1) > 0$. By ergodicity, we also have $\xi_1(x) < 1$ for μ-almost every $x \in X$. Thus, given $k \in \mathbb{N}$, there exists $\delta_k < 1$ such that $E_k = \{x \in X : \xi_1(x) < \delta_k\}$ has measure $\mu(E_k) > 1 - 1/k$. Applying the Birkhoff ergodic theorem to the functions 1_{E_k}, we obtain $B \subseteq X$ of full measure such that for all $x \in B$ and k,

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le i < n \mid \xi_1(g^i(x)) < \delta_k\} = \mu(E_k) > 1 - 1/k. \tag{1.3}$$

By the ergodicity of μ, there exists another set $C \subseteq X$ of full measure whose empirical measures converge to μ, and since $\mathcal{S}f_x \prec 0$ almost everywhere in $x \in X$, there exists a set D of full measure such that $\mathcal{S}f_{g^n(x)} \prec 0$ for all $n \ge 0, x \in D$. Let $x_0 \in A \cap B \cap C \cap D$ and $y_0 \in (\xi_0(x), \xi_1(x))$, and let $\nu \in \mathcal{M}(x, y) \setminus \{\nu_0\}$ be a limit point of the empirical measures different from ν_0. Note that since $x_0 \in B$, necessarily $(\pi_X)_*\nu = \mu$, but since $\nu \neq \nu_0$, it follows that $\nu(X \times \{0\}) < 1$. Thus, $\operatorname{supp} \nu$ is not contained in $X \times \{0\}$, and there exists $\varepsilon > 0$ such that $U = X \times (\varepsilon, 1]$ satisfies $\nu(U) > 0$. Let k be such that $\nu(U) > 1/k$, and $n_j \to \infty$ such that $\liminf_{j\to\infty} e_{n_j}(x_0, y_0)(U) > 1/k$. In other words, we have

$$\liminf_{j\to\infty} \frac{1}{n_j} \#\{0 \le i < n_j \mid f^i_{x_0}(y_0) > \varepsilon\} > 1/k. \tag{1.4}$$

Now, consider the sequence $r_n = \rho(0, f^n_{x_0}(y_0), f^n_{x_0}(\xi_1(x_0)), 1)$, which is well-defined and positive since $0 < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < 1$ by the definition of the points. Note also that the graph of ξ_1 is strictly invariant under F, so $f^n_{x_0}(\xi_1(x_0)) = \xi_1(g^n(x_0))$.

Since $x \in D$, it follows by lemma 2 that r_n is an increasing sequence. On one hand, it is not possible for $r_n \to \infty$, because from eqs. (1.3) and (1.4) we know that there exist $n \in \mathbb{N}$ arbitrarily large such that $0 < \varepsilon < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < \delta_k < 1$. On the other hand, if $r_n \to r < \infty$, then for each $x \in X$ there exists a unique $\xi(x)$ such that $\rho(0, \xi(x), \xi_1(x), 1) = r$. In fact, a calculation shows that

$$\xi(x) = \frac{\xi_1(x)}{r(1 - \xi_1(x)) + \xi_1(x)}.$$

Note that $r_n \to r$ implies $\lim_{n\to\infty}|\xi(g^n(x_0)) - f^n_{x_0}(y_0)| = 0$. Since we know that x_0 is in the basin of μ, necessarily $e_n(x_0, y_0) \Rightarrow \eta$, where $\eta = (\operatorname{graph}\xi)_*\mu$ and $(\operatorname{graph}\xi)(x) = (x, \xi(x))$. Thus, η is an invariant measure. But then,

$$\varphi(x, y) = \frac{\rho(0, y, \xi_1(x), 1)}{1 + \rho(0, y, \xi_1(x), 1)}$$

is an integrable function defined at η-almost every point, with the property that $\varphi(F(x, y)) > \varphi(x, y)$ at η-almost every point. This is impossible, as it would imply $\int \varphi\, d\eta < \int \varphi \circ F\, d\eta = \int \varphi\, d\eta$. □

Example 1.2.2. In example 1.2.1, we have $q_a'(y) = 1 + a(1 - 2y)$, $q_a''(y) = -2a$ and $q_a''' \equiv 0$, so $\mathcal{S}q_a(y) < 0$ whenever $a \neq 0$. Since $q(x) = 0$ only at two points, by theorem 8 there exists $\sigma\colon S^1 \to I$ that separates the two basins.

Positive Schwarzian

If $f\colon M \to M$ is a diffeomorphism on a manifold, we say that an invariant measure μ is *asymptotic* with respect to a reference measure if the empirical measures of almost every point of the reference measure converges to μ. Intuitively, μ describes the entire statistical behavior of the system, up to a set of zero volume.

Since theorem 8 does not require that both basins have positive measure, we also obtain a more general statement than the version in [1, Theorem 5.1], which requires both Lyapunov exponents to be positive. The proof below is also somewhat different, more similar to the proof of theorem 8.

Theorem 9. *If $\mathcal{S}f_x \succ 0$ almost everywhere, then F has an asymptotic measure ν with respect to $\mu \otimes \mathrm{Leb}$. Furthermore, for each $i \in \{0, 1\}$, $\mathrm{Lyap}(\mathcal{A}_i) > 0$ implies $\nu(\mathcal{A}_i) = 0$.*

The idea of the proof is to pass to the invertible extension of the system and observe that the inverses of the fiber maps have negative Schwarzian. By applying theorem 8, we obtain an invariant graph that separates the two basins of the inverse transformation. Since the orbits diverge from the graph for the inverse of the extension, they converge to the graph for the extension. Thus, the graph describes an asymptotic measure for the extension. The pushforward of this measure is the asymptotic measure for the original map.

Proof. Let $(\tilde{X}, \tilde{g}, \tilde{\mu})$ be the invertible extension of (X, g, μ), $\pi\colon (\tilde{X}, \tilde{g}, \tilde{\mu}) \to (X, g, \mu)$ the associated conjugation, and

$$\tilde{F}(\tilde{x}, y) = (\tilde{g}(\tilde{x}), f_{\pi(\tilde{x})}(y)).$$

Note that $\tilde{F}^{-1}(x, y) = (\tilde{g}^{-1}(x), f^{-1}_{\pi(\tilde{g}^{-1}(x))}(y))$. Since $\pi_*\tilde{\mu} = \mu$ and $\tilde{g}_*\tilde{\mu} = \tilde{\mu}$, by proposition 1 we have $f^{-1}_{\pi(\tilde{g}^{-1}(\tilde{x}))} \prec 0$ for almost every $\tilde{x} \in \tilde{X}$. Thus, we can apply theorem 8 to F^{-1} and obtain a function $\sigma\colon \tilde{X} \to I$ that separates the basins $\tilde{\mathcal{B}}_1$ and $\tilde{\mathcal{B}}_0$.

The hyperbolic metric (which coincides with the Hilbert metric) on $(0, 1)$ is given by

$$d_H(r, s) = \log|\rho(0, r, s, 1)|.$$

This metric induces the usual topology of the interval. Since maps with positive Schwarzian decrease cross-ratios, the condition $\mathcal{S}f_x \succ 0$ means that f_x is a contraction (not necessarily uniform) for this metric.

Let $A \subseteq \tilde{X}$ be a set of full $\tilde{\mu}$ measure where $\mathcal{S}f_{\pi(\tilde{g}^n(\tilde{x}))} \succ 0$ for all $n \geq 0, \tilde{x} \in A$, and $B \subseteq \tilde{X}$ another set of full measure where the empirical measures of $\tilde{x} \in B$ converge

to the ergodic measure $\tilde{\mu}$. It follows that for all $\tilde{x} \in A \cap B$ and $y \in (0,1)$, the limit $r = \lim_{n\to\infty} d_H(f^n_{\pi(\tilde{x})}(y), \sigma(\tilde{g}^n(\tilde{x})))$ exists and is zero if and only if the orbit of $(\tilde{x}, y)$ converges in distance to the graph of σ under $\tilde{F}$, which in turn implies $e_n(\tilde{x}, y) \Rightarrow \tilde{\nu} = (\operatorname{graph}\sigma)_*\tilde{\mu}$.

Suppose $r > 0$. Then, as in the previous proof, there would exist another graph ξ, different from σ almost everywhere, such that $e_n(\tilde{x}, y) \Rightarrow \eta = (\operatorname{graph}\xi)_*\tilde{\mu}$. Since $\varphi(\tilde{x}, y) = d_H(y, \sigma(\tilde{x}))$ is such that $\varphi(F(\tilde{x}, y)) < \varphi(\tilde{x}, y)$ whenever $(\tilde{x}, y) \in \operatorname{graph}\xi$ and η is an invariant measure for F with $\eta(\operatorname{graph}\xi) = 1$, we reach a contradiction.

This shows that $E = (A \cap B) \times (0,1)$ is a set of full $\tilde{\mu} \otimes \mathrm{Leb}$ measure such that $(\tilde{x}, y) \in E$ implies $e_n(\tilde{x}, y) \Rightarrow \tilde{\nu}$. Let $\nu = \pi_*\tilde{\nu} \in \mathcal{M}(X \times I)$. The measure ν is invariant, since

$$F_*\nu = (F \circ \pi)_*\tilde{\nu} = (\pi \circ \tilde{F})_*\tilde{\nu} = \pi_*\tilde{\nu} = \nu.$$

Furthermore, the basin of ν has full Lebesgue measure. In fact, if $\varphi \in C(X \times I)$, then $\varphi \circ \pi \in C(\tilde{X} \times I)$ and $e_n(\pi(\tilde{x}), y)(\varphi) = e_n(\tilde{x}, y)(\varphi \circ \pi)$. Therefore, $e_n(x, y) \to \nu$ if and only if $e_n(\tilde{x}, y) \to \tilde{\nu}$ for any $\tilde{x} \in \pi^{-1}(x)$, that is, $(\pi \times \mathrm{id})^{-1}(\mathcal{B}(\nu)) = \mathcal{B}(\tilde{\nu})$. It follows that $(\mu \otimes \mathrm{Leb})(\mathcal{B}(\nu)) = (\tilde{\mu} \otimes \mathrm{Leb})(\mathcal{B}(\tilde{\nu})) = 1$. □

Remark 1. The proofs of theorems 8 and 9 would carry out mostly unchanged if instead of assuming negative Schwarzian for the fiber maps, we assumed that one basin has positive measure and that for every x there is a metric d_x in I, equivalent and uniformly stronger than the Euclidian metric, with the property that $d_x(y_1, y_2) < d_{g(x)}(f_x(y_1), f_x(y_2))$. In the negative Schwarzian case, this metric is the hyperbolic metric.

Chapter 2

Historical behavior

Ergodic theory tries to find order in chaos by hoping that even if a dynamical system looks complicated, its statistical properties are well behaved. But what if the statistics don't even exist for most points? This phenomenon is what is called *non-statistical*, or *historical* behavior.

2.1 Martingale-coboundary decomposition

rewrite this section for ergodic processes in general, I think it will be less confusing later (see the remark in the next theorem)

The key idea for applying theorem 22 in a dynamical system $(X, \mathcal{F}, \mu, f)$, where μ is a f-invariant probability, is finding a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ for which $\varphi \in L^2(\mu)$ can be decomposed as a sum

$$\varphi = \varphi_0 + \phi \circ f - \phi \tag{2.1}$$

such that $\phi \in L^2(\mu)$, φ_0 is $\mathcal{G}$-measurable and $\eta = (\varphi_0 \circ f^{i-1})_{i\geq 1}$ is a reversed martingale difference sequence with respect to $(f^{-i}(\mathcal{G}))$. Indeed, this allows the transference of the weak invariant principle for the original φ: let $\xi = (\varphi \circ f^{i-1})_{i\geq 1}$, and notice that

$$S_n^\xi = S_n^\eta + \phi \circ f^n - \phi,$$

so

$$D_n^\xi(x)(t) = \frac{S^\xi_{\lfloor nt \rfloor}(x)}{\sqrt{n}} = D_n^\eta(x)(t) + \frac{1}{\sqrt{n}}\big(\phi \circ f^{\lfloor nt \rfloor} - \phi\big)(x).$$

It suffices to show that for μ-almost every x, the second term in the sum converges uniformly to zero (as a function of t). Given $\varepsilon > 0$, we look at the sets $B_n = \{x : (\phi^2 \circ f^n)(x) \geq n\varepsilon\}$. Since

$$\sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(\{\phi^2 \geq n\varepsilon\}) = \int \lfloor \phi^2/\varepsilon \rfloor \, d\mu \leq \frac{1}{\varepsilon}\|\phi\|_2^2,$$

by Borel-Cantelli, $\mu(\limsup_{n\to\infty} B_n) = 0$. If we take the intersection over k of the sets $\liminf_{n\to\infty} B_n^c$ for $\varepsilon = 1/k$, we get a full-measure set G with the property that for every $x \in G$ and $\varepsilon > 0$, there exists N such that $n \geq N$ implies $\frac{1}{\sqrt{n}}|\phi(f^n(x))| \leq \varepsilon$. Now, let $M \geq N$ be big enough so that $n \geq M$ implies $\frac{1}{\sqrt{n}}|\phi(f^i(x))| \leq \varepsilon$ for all $0 \leq i \leq N$. So for $n \geq M$,

$$\sup_{t\in[0,1]} \frac{1}{\sqrt{n}}\big|\phi \circ f^{\lfloor nt \rfloor} - \phi\big|(x) \leq \max_{0\leq i\leq N} \frac{|\phi \circ f^i(x)|}{\sqrt{n}} + \max_{N\leq i\leq n} \frac{|\phi \circ f^i(x)|}{\sqrt{i}} \leq 2\varepsilon$$

as desired.

Existence of martingale-coboundary decomposition

From now on, we assume that μ is an ergodic measure for f.

Theorem 10. *Suppose $\varphi \in L^2(\mu)$ and $\mathcal{F}_j = f^{-j}(\mathcal{F})$ are such that*

$$\sum_{n=0}^{\infty} \|\mathbb{E}_\mu(\varphi \mid \mathcal{F}_j)\|_2 < \infty.$$

Then a decomposition as in eq. (2.1) exists. Moreover, this implies that

$$\int \varphi \, d\mu = \int \varphi_0 \, d\mu = 0$$

and that the series in

$$\sigma^2 = \|\varphi_0\|_2^2 = \|\varphi\|_2^2 + 2\sum_{n=1}^{\infty} \int \varphi \cdot \varphi \circ f^n \, d\mu$$

converges absolutely. By the preceding arguments, if $\varphi_0 \neq 0$ then $(D_n^\xi)_* \mu \Rightarrow \sigma W$.

Proof. We begin by writing $\phi = \phi^- + \phi^+$, where $\phi^+ = \mathbb{E}_\mu(\phi \mid \mathcal{G}_0)$.

write □

Remark 2. If φ was a coboundary $\varphi = \phi \circ f - \phi$ to begin with, then it is clear that $S_n^\xi = \phi \circ f^n - \phi$, so the iterates of the skew product $F(x, y) = (f(x), y + \varphi(x))$ satisfy

$$F^n(x_0, y_0) = (f^n(x_0), \phi(f^n(x_0)) - \phi(x_0) + y_0) \subseteq \text{graph}(\psi)$$

and are thus contained in the graph of ψ, where $\psi(x) = \phi(x) - \phi(x_0) + y_0$. In particular, F is not ergodic for $\mu \otimes \text{Leb}$.

Corollary 11. *Suppose φ is such that there exists a summable sequence $c_n > 0$ satisfying*

$$\int (\psi \circ f^n)\varphi \, d\mu \leq \|\psi\|_2 c_n$$

for all $\psi \in L^2(\mathcal{G})$, where $\mathcal{G}$ is as in theorem 10. Then

$$\sum_{n=0}^{\infty} \|\mathbb{E}(\varphi \mid f^{-n}(\mathcal{G}))\|_2 < \infty.$$

Remark 3. The book [2] proves this condition is satisfied for smooth expanding maps, Hölder-continuous φ and $\mathcal{G} = \mathcal{F}$, so this includes the $f(x) = kx$ maps. The second condition is trivial when $\mathcal{G} = \mathcal{F}$.

Remark 4. More generally, [2] also shows the decomposition (2.1) exists for uniformly hyperbolic f and Hölder φ, also by showing that a sum like in theorem 10 converges due to decay of correlations, then uses this to show the central limit theorem (but I have not read the proof). In this case, $\mathcal{G}$ has something to do with the attractor and the second condition follows from Hölder somehow.

the examples in the Billingsley book are worth checking. They include both the kx maps (hmm... I guess B-M didn't see it?) and the Gauss map.

2.2 Historical behavior for skew products

Let (X, μ) be a Borel probability space, where X is compact and metrizable, and $F\colon X \times I \to X \times I$ a skew product of the form

$$F(x, y) = (g(x), f_x(y))$$

where $g\colon X \to X$ is ergodic and measure-preserving. Suppose there is a homeomorphism $\gamma\colon I \to \overline{\mathbb{R}}$ and a measurable function $p\colon X \to \mathbb{R}$ satisfying $t_x \circ \gamma = \gamma \circ f_x$ for all $x \in X$, where $t_x(y) = y + p(x)$ is the translation by $p(x)$.

$$\begin{tikzcd} I \arrow[r, "f_x"] \arrow[d, "\gamma"] & I \arrow[d, "\gamma"] \\ \overline{\mathbb{R}} \arrow[r, "t_x"] & \overline{\mathbb{R}} \end{tikzcd}$$

Note there is no loss of generality in assuming $\gamma(0) = -\infty$, $\gamma(1) = +\infty$ and $\gamma\left(\frac{1}{2}\right) = 0$.

Theorem 12. *If the weak invariance principle is valid for the pair (g, p), in the sense $(D_n^\xi)_*\mu \Rightarrow \sigma W$ for $\sigma > 0$ and $\xi = (p \circ g^{i-1})_{i \geq 1}$, then F has historical behavior. In fact, for μ-almost every $x \in X$ and every $y \in (0, 1)$,*

$$\{e_n(x, y)\}'_{n \in \mathbb{N}} = [\delta_-, \delta_+] \tag{2.2}$$

where $\delta_- = \mu \otimes \delta_0$ and $\delta_+ = \mu \otimes \delta_1$ and the limit points are taken in the weak topology.*

Proof. Fix $\delta \in (0, 1)$, and define

$$\begin{aligned} T(x, y) &= \limsup_{n \to \infty} \frac{1}{n} \#\{0 \leq i < n \mid f_x^{(i)}(y) > \delta\} \\ &= \limsup_{n \to \infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{(M_y, \infty)}(S_i^\xi(x)) \end{aligned}$$

where $M_y = \gamma(\delta) - \gamma(y)$. Let E be the full μ-measure set given by lemma 21.

Claim 1: $T(x, y_1) = T(x, y_2)$ for all $x \in E$, $y_1, y_2 \in (0, 1)$.

Indeed, set $M = \max\{|M_{y_1}|, |M_{y_2}|\}$ and let $G = \{i : |S_i^\xi(x)| > M\}$. Since $1_{(M_{y_1}, \infty)}(t) = 1_{(M_{y_2}, \infty)}(t)$ whenever $|t| > M$, for all $i \in G$,

$$1_{(M_{y_1}, \infty)}(S_i^\xi(x)) = 1_{(M_{y_2}, \infty)}(S_i^\xi(x)).$$

So

$$\begin{aligned} |T(x, y_1) - T(x, y_2)| &\leq \limsup_{n \to \infty} \frac{1}{n} \sum_{i=0}^{n-1} |1_{(M_{y_1}, \infty)}(S_i^\xi(x)) - 1_{(M_{y_2}, \infty)}(S_i^\xi(x))| \\ &\leq \limsup_{n \to \infty} \frac{1}{n} \#\{0 \leq i < n \mid i \notin G\} = 0. \end{aligned}$$

This proves the claim. Since T is invariant by F, for any $x \in g^{-1}(E)$,

$$T(x, y) = T(g(x), f_x(y)) = T(g(x), y).$$

By ergodicity, there is a full measure set $H \subseteq X$ and $t \in [0,1]$ for which $T(x,y) = t$ for all $x \in H$, $y \in (0,1)$.

We can use the arcsine law to evaluate this constant. Given $\varepsilon > 0$, fix some $y \in (0,1)$ and define

$$E_n = \left\{ x \in X \;\middle|\; \frac{1}{n}\#\{0 \le i < n \mid S_i^{\xi}(x) > M_y\} > 1 - \varepsilon \right\}.$$

By the lemma 20, we know $\lim_n \mu(E_n) = Z([1-\varepsilon, 1]) > 0$, so

$$\mu(\limsup_{n\to\infty} E_n) \ge \limsup_{n\to\infty} \mu(E_n) > 0.$$

This means there exists $x_0 \in H \cap \limsup E_n$, and since

$$\limsup_{n\to\infty} E_n = \{x \in X \mid T(x,y) > 1 - \varepsilon\}$$

we have $t = T(x_0, y) > 1 - \varepsilon$. But $\varepsilon > 0$ is arbitrary, so $T(x,y) = 1$ for every $x \in H$ and $y \in (0,1)$.

So far, we started with $\delta \in (0,1)$ and showed that for

$$T_\delta(x,y) = \limsup_{n\to\infty} \frac{1}{n}\#\{0 \le i < n \mid f_x^{(i)}(y) > \delta\},$$

there is $H_\delta \subseteq X$ with $\mu(H_\delta) = 1$ such that for $x \in H_\delta$, $y \in (0,1)$, $H_\delta(x,y) = 1$. But note that for fixed x, y, the function $\delta \mapsto T_\delta(x,y)$ is monotone. With this observation, it's easy to see that if we define

$$H = \bigcap_{i=1}^{\infty} H_{1/i} \cap H_{1-1/i},$$

then $\mu(F) = 1$ and $T_\delta(x,y) = 1$ for all $x \in H$ and $y, \delta \in (0,1)$.

In particular, this implies that if $x \in H$ and $y \in (0,1)$, then for any $\delta \in (0,1)$, letting $U_n = \{0 \le i < n \mid f_x^{(i)}(y) > \delta\}$, the limit supremum of the y coordinate satisfies

$$\limsup_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} f_x^{(i)}(y) \ge \limsup_{n\to\infty} \frac{1}{n}\sum_{i\in U_n} \delta = \delta,$$

so the left-hand side is equal to 1. The proof we have so far works analogously for the time averages of the iterates in which $f_x^{(i)}(y) < \delta$, and by doing so we get

$$\liminf_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} f_x^{(i)}(y) = 0.$$

Note that those are time averages of the continuous bounded function $\pi_2 \colon X \times I \to \mathbb{R}$, so historical behavior is proved.

We now focus on eq. (2.2). By Birkhoff, we can fix a full measure set $S \subseteq X$ such that for all $x \in S$ and $\varphi \in C(X)$,

$$\frac{1}{n}\sum_{i=0}^{n-1} \varphi(g^i(x)) \to \int_X \varphi \, d\mu.$$

Given x in the full measure set $H \cap S$ and $y \in (0,1)$, we construct an increasing sequence $(n_k^-)_k \subseteq \mathbb{N}$ inductively, letting $n_0^- = 1$ and $n_{k+1}^- > n_k^+$ be any integer satisfying

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid f_x^{(i)}(y) < 1/k\} > 1 - 1/k.$$

Analogously, we can pick an increasing sequence $(n_k^+)_k \subseteq \mathbb{N}$ for which

$$\frac{1}{n_k^+}\#\{0 \le i < n_k^+ \mid f_x^{(i)}(y) > 1 - 1/k\} > 1 - 1/k.$$

Claim 2:

$$e_{n_k^-}(x,y) \Rightarrow \delta_- \quad \text{and} \quad e_{n_k^+}(x,y) \Rightarrow \delta_+.$$

We only show the first convergence, since the second is analogous.

Let $\phi \in C(X \times I)$ and $\varepsilon > 0$. Since ϕ is uniformly continuous, there is some k_1 for which

$$\sup\big\{|\phi(x,1) - \phi(x,s)| : x \in X, s \in [0, 1/k_1]\big\} < \varepsilon.$$

Let $D = \{i \mid f_x^{(i)}(y) < 1/k_1\}$, and let N be big enough so that for $k \ge N$,

$$\left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,0)) - \int_X \phi(z,0)\, d\mu(z)\right| < \varepsilon$$

(which is possible since $x \in S$ and $\varphi(x) = \phi(x,0)$ is continuous), and

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid i \notin D\} < \varepsilon/\|\phi\|_\infty,$$

so we can estimate

$$\begin{aligned}
&|e_{n_k^-}(x,y)(\phi) - \delta_-(\phi)| \\
&\qquad = \left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,y)) - \int_X \phi(z,0)\, d\mu(z)\right| \\
&\qquad \le \left|\frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,y)) - \frac{1}{n_k^-}\sum_{i=0}^{n_k^- - 1}\phi(F^i(x,0))\right| + \varepsilon \\
&\qquad \le \frac{1}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \in D}}\big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + \frac{2}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \notin D}}\|\phi\|_\infty + \varepsilon \\
&\qquad \le \frac{1}{n_k^-}\sum_{\substack{0 \le i < n_k^- \\ i \in D}}\big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + 3\varepsilon \\
&\qquad \le \varepsilon + 3\varepsilon = 4\varepsilon.
\end{aligned}$$

This completes the proof of the claim. To finish the proof of (2.2), we first show the following.

Claim 3: If $\eta \in \mathcal{M}(x,y)$, then $\operatorname{supp}\eta \subseteq X \times \{0,1\}$ and $\eta \in [\delta_0, \delta_1]$.

We use lemma 21 again for this. Let $e_{n_k}(x,y) \Rightarrow \eta$ be a limit point, $x_0 \in X$ and $y_0 \in (0,1)$, so there is a neighborhood U of (x_0, y_0) such that $\overline{U} \subseteq X \times (0,1)$. By the lemma, $e_{n_k}(x,y)(U) \to 0$, so by Portmanteau

$$\eta(U) \le \liminf_{k\to\infty} e_{n_k}(x,y)(U) = 0.$$

This shows $(x_0, y_0) \notin \operatorname{supp}\eta$.

We know η is a limiting measure for x, so it's invariant by F, and the fact $x \in X$ implies $(\pi_1)_*\eta = \mu$. This means that $\eta \ll \delta_- + \delta_+$, and since δ_-, δ_+ are ergodic measures, η must be a convex combination of the two.

We have shown $\mathcal{M}(x, y) \subseteq [\delta_-, \delta_+]$. To conclude eq. (2.2), it suffices to note that by proposition 25, $\mathcal{M}(x, y)$ is connected, and since it contains both δ_- and δ_+, it must be the whole segment. □

Remark 5 (A different proof). Another way to show $\delta_-, \delta_+ \in \mathcal{M}(x)$ for x in a full measure set H is as follows: let $\delta > 0$ and $p_\delta \colon [0, 1] \to \mathbb{R}$ be any continuous function such that $p_\delta(0) = 0$ and $p(t) > 1 + \delta$ for $t > \delta$. Let $U_\delta = B_\delta(p_\delta)$. By the Forgery theorem we know $W(U_\delta) > 0$,

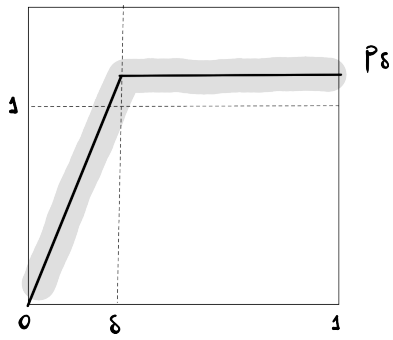

Figure 2.1: The function p_δ

so by theorem 14

$$\liminf_{n\to\infty}((D_n^\xi)_*\mu)(U_\delta) \geq \sigma W(U_\delta) > 0.$$

Therefore,

$$S_\delta = \limsup_{n\to\infty} (D_n^\xi)^{-1}(U_\delta) \subseteq X$$

has positive measure. This set is also invariant for g because

$$\begin{aligned}
g^{-1}(S_\delta) &= \limsup_{n\to\infty}\{x \in X \mid D_n^\xi(g(x)) \in U_\delta\} \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt \rfloor + 1}(x) - p(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt' \rfloor}(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} \text{ where } t' = t + 1/n \\
&= \left\{ x \in X \;\middle|\; \liminf_{n\to\infty} \sup_t \left| \frac{S^\xi_{\lfloor nt \rfloor}(x)}{\sqrt{n}} - p_\delta(t) \right| < \delta \right\} = S_\delta.
\end{aligned}$$

Therefore $\mu(S_\delta) = 1$. Now, note that if $x \in S_\delta$, then there is $n_k^+ \to \infty$ for which for every k, $D^\xi_{n_k}(x) \in U_\delta$. This in turn implies that

$$\#\left\{ 0 \leq i < n_k^+ \;\middle|\; S_i^\xi(x) > \sqrt{n_k^+} \right\} \geq (1 - \delta) n_k^+.$$

Analogously, there is $n_k^- \to \infty$ such that for every k,

$$\#\left\{ 0 \leq i < n_k^- \;\middle|\; S_i^\xi(x) < -\sqrt{n_k^-} \right\} \geq (1 - \delta) n_k^-.$$

Both can be used just as in the previous proof.

Theorem 13. *Under the same hypothesis as theorem 12, for every* $y \in (0,1)$,

$$(e_n)_*(\mu \otimes \delta_y) \Rightarrow \delta_* Z \tag{2.3}$$

where Z *is the arcsine measure on* $[0,1]$ *and* $\delta(\lambda) = \lambda\delta_- + (1-\lambda)\delta_+$.

Proof. Let $F \subseteq \mathcal{M}(X \times I)$ be a closed set, and let $I = \delta([a,b]) = F \cap [\delta_-, \delta_+]$.

Idea: first, show that in I we get the desired statistics; then, show that in $F \setminus I$ the times converge to zero. □

Question 1. Is the statement above true for $\lambda \otimes \delta_y$ if $\lambda \ll \mu$?

Question 2. Douglas question: is there a measure $\nu \in [\delta_0, \delta_1]$ such that $(F^n)_*\eta \to \nu$ for "nice" η absolutely continuous with respect to $\mu \otimes m$?

Appendix A

Some categorical language

Perhaps the main charm of Category Theory is that of framing definitions and results throughout mathematics in a language that is both general and precise. For this reason I like to learn it as a hobby, and have chosen to state many of the results in these appendices using it. My goal is not to make the text harder to understand, but instead, to give a more conceptual point of view that (understandably) is not commonly found in the dynamical systems literature, but which I personally find interesting and rich. Some benefits from this approach will be scattered around the text. Among them, I can highlight the use formal diagrams (*string diagrams*) to reason about probability models known as Markov processes (in **??**).

Nowadays category theory has made its way into many applied fields, most remarkably computer science in the form of programming languages.

Appendix B

Measure and probability

refs

1. V. I. Bogachev, *Measure Theory*. Berlin, Heidelberg: Springer, 2007
2. E. S. Andersen, "On the introduction of measures in infinite product sets," *Matematisk-Fysiske Meddelelser*, vol. XXV, no. 4,

B.1 The category of probability spaces

Products and limits

We can construct a product measure

B.2 Random variables

Independence and distribution

B.3 Weak convergence

Theorem 14 (Portmanteau)**.** *The following are equivalent:*

i. $\mu_n \Rightarrow \mu$*;*

ii. $\lim_{n\to\infty} \mu_n(\phi) = \mu(\phi)$ *for all Lipschitz functions* ϕ*;*

iii. $\liminf_{n\to\infty} \mu_n(U) \geq \mu(U)$ *for all open sets* U*;*

iv. $\limsup_{n\to\infty} \mu_n(F) \leq \mu(F)$ *for all closed sets* F*;*

v. $\lim_{n\to\infty} \mu_n(A) \leq \mu(A)$ *for all continuity sets* A *of the measure* μ*.*

B.4 Ergodic theorems

Hewitt-Savage zero-one law

B.5 Brownian motion

The space of càdlàg functions

The space $C([0,1])$ of continuous functions does not let us talk about processes that contain discontinuities, or even how discontinuous processes can converge in a suitable way to continuous

ones. This is important in our setting to deal with the convergence of certain processes induced by discrete dynamics to the Brownian motion, which is continuous almost surely.

For this, we shall introduce the space of *càdlag* functions, or Skorohod space. This is the space of functions whose discontinuities are of the *first kind*, that is, both lateral limits exist, and whose value at the point is equal to the right limit:

$$\mathbb{D}([0,1]) = \left\{ f\colon [0,1] \to \mathbb{R} \;\middle|\; \forall t,\ \lim_{s\to t^-} x(s) \text{ exists and } \lim_{s\to t^+} x(s) = x(t) \right\}.$$

We want to put a metric on this space with the property that functions are close together if their values are close to each other uniformly, but also allowing small perturbations in the parametrization of time.

Definition 5 (Skorohod topology)**.** Let Λ be the set of strictly increasing continuous functions from $[0,1]$ to itself. For $f, g \in \mathbb{D}([0,1])$, we define

$$d(f,g) = \inf_{\lambda\in\Lambda} \max\left\{ \|\lambda - \mathrm{id}\|_\infty, \|f - g\circ\lambda\|_\infty \right\}.$$

The distance above defines a metric on $\mathbb{D}([0,1])$, whose topology is called the Skorohod topology. Since $d(f,g) \le \|f-g\|_\infty$, the Skorohod topology is coarser than the uniform topology.

Proposition 15. *The restriction of the Skorohod topology to $C([0,1])$ coincides with the uniform topology.*

Proof. Since the Skorohod topology is coarser, it remains to show that if $f, f_n \in C([0,1])$ and $d(f_n, f) \to 0$, then $\|f_n - f\|_\infty \to 0$. But note

$$\|f_n - f\|_\infty \le \inf_{\lambda\in\Lambda} [\|f_n - f\circ\lambda\|_\infty + \|f\circ\lambda - f\|_\infty] \to 0$$

by the convergence assumption and uniform continuity of f. □

The Wiener measure and arcsine law

insert properties (and construction?) of the Wiener measure $W \in \mathcal{M}_1(\mathbb{D}([0,1]))$.

Let $\tau^+\colon \mathbb{D}([0,1]) \to \mathbb{R}$ be the observable that returns the proportion of time its argument is positive, that is, $\tau^+(f) = \mathrm{Leb}\{x \in [0,1] : f(x) > 0\}$.

Lemma 16. *τ^+ is measurable and continuous in a full W-measure set.*

Proof. adapt this to $\mathbb{D}([0,1])$ This is proved in [3, appendix M15]. Since the normal distributions are non-atomic and for every $t \in [0,1]$, W_t has normal distribution, $W(\{f \in \mathbb{D}([0,1]) : f(t) = 0\}) = 0$ for all t, so by Fubini, $\mathrm{Leb}(\{t \in [0,1] : f(t) = 0\}) = 0$ for W-almost every f. It remains to check that τ^+ is continuous at such points $f \in W$. Indeed, 0 is the only discontinuity of the indicator function $1_{(0,\infty)}$, so if f is in this set and $f_n \to f$ uniformly, then $1_{(0,\infty)} \circ f_n \to 1_{(0,\infty)} \circ f$ almost everywhere, so by the DCT the same is true of their respective integrals. □

Theorem 17 (Arcsine law)**.** *For $\alpha \in [0,1]$,*

$$(\tau^+)_* W([0,\alpha]) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

Theorem 18 (Forgery theorem)**.** *If $f \in C([0,1])$ and $f(0) = 0$, then every neighborhood of f has positive Wiener measure.*

Weak invariance principles

We will fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for the rest of this section. Given a sequence $\xi = (\xi_1, \xi_2, \dots)$ of random variables defined on Ω, we can define their partial sums $S_n^\xi = \sum_{i=1}^n \xi_i$ and a sequence of processes $D_n^\xi \colon \Omega \to \mathbb{D}([0,1])$ given by

$$D_n^\xi(\omega)(t) = \frac{S_{\lfloor nt \rfloor}^\xi(\omega)}{\sqrt{n}},$$

obtained by appropriately scaling the partial sums of the random variables up to some iterate n.

Remarkably, for many classes of sequences ξ, their corresponding processes D_n^ξ converge in some way to the Brownian motion — we denote by an *invariance principle* a result of this form. Perhaps the best known invariance principle is the Donsker's theorem, which is a *weak* invariance principle since it asserts convergence in the weak topology.

Theorem 19 (Donsker)**.** *If $\xi \subseteq L^2(\mathbb{P})$ is a sequence of i.i.d. variables with mean zero and variance 1, then the measures $(D_n^\xi)_*\mathbb{P}$ converge weakly to the Wiener measure. In other words, for every measurable continuous function $\varphi \colon \mathbb{D}([0,1]) \to \mathbb{R}$,*

$$\int \varphi \circ D_n^\xi \, d\mathbb{P} \to \int \varphi \, dW.$$

The motivation for looking into invariance principles is that they allow transferring certain results from Brownian motions to the sequence of partial sums S_n^ξ. In our case, the result we are interested in is the arcsine law, and since we know from lemma 16 that τ^+ is continuous a.e., weak convergence is enough:

Lemma 20. *Suppose $\xi \subseteq L^2(\mathbb{P})$ is such that $(D_n^\xi)_*\mathbb{P} \Rightarrow W$. Then the average of the time in which the random walk S_n^ξ is above $M \in \mathbb{R}$ converges to an arcsine distribution, in the sense that*

$$\lim_{n\to\infty} \mathbb{P}\left(\left\{ \omega \in \Omega \;\middle|\; \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\} \le \alpha \right\} \right) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

Proof. Writing $\tau_n^+(\omega) = \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\}$, the left-hand side can be rewritten as $\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha])$. Set $c_n = M/\sqrt{n}$, and note that

$$\tau_n^+(\omega) = \frac{1}{n}\#\left\{ 1 \le i \le n \;\middle|\; \frac{S_i^\xi(\omega)}{\sqrt{n}} - \frac{M}{\sqrt{n}} > 0 \right\} = \tau^+(D_n^\xi(\omega) - c_n).$$

Since by theorem 17 we know $Z = (\tau^+)_*W$ is equivalent to the Lebesgue measure on $[0,1]$, intervals are continuity sets for it. By the weak convergence assumption and continuity of τ^+, we know $(\tau^+ \circ D_n^\xi)_*\mathbb{P} \Rightarrow Z$, and since $c_n \to 0$ (as a sequence in $\mathbb{D}([0,1])$), in fact $(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P} \Rightarrow Z$. By Portmanteau, this means

$$\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha]) = \lim_{n\to\infty}(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P}([0,\alpha]) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

□

Lemma 21. *Under the same hypothesis as lemma 20, for all $M \in \mathbb{R}$ and for $\mathbb{P}$-almost every $\omega \in \Omega$,*

$$\lim_{n\to\infty} \frac{1}{n}\#\{1 \le i \le n \mid S_i^\xi(\omega) \in [-M, M]\} = 0.$$

Proof. Just like in the previous proof, write $\tau_n(\omega) = \frac{1}{n}\#\{1 \le i \le n : \mid S_n^\xi(\omega) \in [-M, M]\}$, and then note

$$\tau_n(\omega) = \tau^+(D_n^\xi(\omega) + c_n) - \tau^+(D_n^\xi(\omega) - c_n).$$

where $c_n = M/\sqrt{n}$. So

$$\lim \int \tau_n \, d\mathbb{P} = \int \tau^+ \, dW - \int \tau^+ \, dW = 0.$$

I haven't figured out how to continue from here, or if the statement is true. (I think it certainly is) □

Ergodic Martingale Differences

Donsker's theorem is far from being the most general result of its kind. In fact, it can be generalized to a weak invariance principle for ergodic martingale difference sequences, as stated below [see 3, theorem 18.3].

Definition 6. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables taking values in Y is said to be *stationary* if $\xi_*\mathbb{P}$ is invariant by the left shift $\sigma \colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$.

Definition 7. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be *ergodic* if $\xi_*\mathbb{P}$ is ergodic for the left shift.

Definition 8. Every sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables generates a filtration $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ of $\mathcal{F}$. We say that ξ is a *martingale difference sequence* if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \mathcal{F}_{n-1}) = 0.$$

Theorem 22. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic martingale difference sequence for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^\xi)_*\mathbb{P} \Rightarrow \sigma W$.*

Reverse martingale difference sequences

In ergodic dynamics, the processes we usually care about are the values $\xi = (\varphi \circ f^i)_{i \ge 0}$ of some observable $\varphi \in L^1(\mu)$ along orbits of f, especially if f is not invertible. This process, however, cannot possibly be a martingale difference sequence unless $\varphi = 0$ because $\varphi \circ f^i$ is measurable for the σ-algebra $f^{-i}(\mathcal{F})$, which is non-increasing. Nevertheless, if μ is a stationary measure for f, we can always look into the invertible extension to get an associated process which satisfy the martingale conditions.

Consider the space $\mathbb{R}^{\mathbb{N}}$ with the product σ-algebra $\mathcal{H} = \mathcal{B}^{\otimes \mathbb{N}}$ and the measure $\nu = \xi_*\mu$. This measure is invariant for the left shift $\sigma \colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ since $\sigma \circ \xi = (\varphi \circ f^{i+1})_{i \ge 0} = \sigma \circ f$, so $\sigma_*\nu = (\xi \circ f)_*\mu = \nu$. Since $\mathcal{H}$ coincides with the Borel σ-algebra, the construction of the invertible extension (see appendix C.1) gives us a stationary measure $\tilde{\nu} \in \mathcal{M}(\mathbb{R}^{\mathbb{Z}})$ for the invertible shift, which we shall call $\tilde{\sigma}$, whose projection $\pi_*\tilde{\nu}$ by $\pi \colon \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{N}}$ is equal to ν. The measure $\tilde{\nu}$ is ergodic if and only if ν is.

We can now consider the inverted process $\tilde{\xi} = (\pi_{-i})_{i \ge 0}$ with respect to this new probability space $(\mathbb{R}^{\mathbb{Z}}, \tilde{\nu})$, and $\tilde{\xi}$ will be a martingale difference sequence if

$$\mathbb{E}_{\tilde{\nu}}(\pi_{-n} \mid \mathcal{B}_{-n+1}) = 0.$$

Since $\tilde{\sigma}$ is surjective, this is equivalent to

$$\mathbb{E}_{\tilde{\nu}}(\pi_0 \mid \mathcal{B}_1) = 0,$$

which ultimately translates to $\mathbb{E}_\mu(\varphi \mid f^{-1}(\mathcal{F})) = 0$. Whenever this happens, we say ξ is a *reverse martingale difference sequence*. If μ is ergodic for f, since this implies that $\tilde{\xi}$ is a stationary ergodic martingale difference sequence, we get theorem 22 for $\tilde{\xi}$. This, in turn, also implies the weak invariance principle for the original ξ, since due to stationarity, $(D_n^{\tilde{\xi}})_*\tilde{\nu} = (D_n^{\tilde{\xi}} \circ \tilde{\sigma}^n)_*\tilde{\nu} = (D_n^{\xi})_*\mu$.

Corollary 23. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic reverse martingale difference sequence for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^{\xi})_*\mathbb{P} \Rightarrow \sigma W$.*

Appendix C

Ergodic theory and dynamics

C.1 Categorical notions

The starting point of the field of dynamical systems is a space X, endowed with some structure, along with some notion of "time evolution", which could be a map $X \to X$ or a family thereof, which preserves this structure. For instance, in topological dynamics X is usually a topological space and the time evolution is given by a continuous map, while in ergodic theory the space is endowed with the structure of a measure and the maps are measure-preserving.

These different notions can be encompassed by the notion of a *monoid object* in a category. Following [6], we can define a monoid object in a category $\mathcal{C}$ as a functor $M \to \mathcal{C}$, where M is *monoid*, that is, a category with a single object. Let's recall that an usual monoid is like a group but without the inverse operation, that is, a set M endowed with a special identity element 1_M and a multiplication $(a, b) \in X \times X \mapsto a \cdot b \in X$ satisfying

i. $1_M \cdot a = a = a \cdot 1_M$ for all $a \in M$;

ii. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in M$.

It's readily seen that given such a monoid M, the category composed of a single object $\bigstar$, set of morphisms $\mathrm{Hom}(\bigstar, \bigstar) = M$ and composition given by the monoid multiplication will satisfy the definition of a category.

To illustrate the definition, if $f_a, f_b \colon X \to X$ are two measure-preserving maps, then the associated monoid is the one-object category for the free monoid with two generators $\{a, b\}$, and the functor sends $\bigstar$ to the object X and words to the compositions of f_a and f_b in the category of measure-preserving maps, e.g., *abba* is sent to $f_a \circ f_b \circ f_b \circ f_a \colon X \to X$.

This point of view also allows for situations in which the morphisms carry more structure than a single function from a set to itself satisfying some properties. A great example of this are Markov chains, whose time evolution does not map points to points but to distributions. Another example is the action of an iterated function system on closed subsets of a compact metric space, whose time evolution maps points to a closed subset — namely, the set of places where it could be if we pick one of the maps at random. Below, we will describe both briefly with more details.

Some examples

Category of Markov kernels

Sometimes, it's useful to model uncertainty in dynamics by having the maps send each point $x \in X$ not to a point $f(x) \in X$, but to a *probability measure* $f(x) \in \mathcal{M}_1(X)$. The probability

measure $f(x)$ then encodes the distribution of expected outcomes given that we started in the state x.

It may not be so clear, however, how we can compose iterates of f and regard it as a dynamic on X. Semantically, we want the iterates to encode the distribution of expected outcomes after n steps given that the system started at the point x. To work it out, we can start with the case where X is finite, so any $f(x) \in \mathcal{M}_1(X)$ is completely determined by some transition probabilities $\mathbb{P}(f(x) = y) := f(x)(\{y\})$. Then, supposing that the probability of first making the transition $x \mapsto y$ and then $y \mapsto z$ is $\mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z)$, we should have

$$\mathbb{P}(f^2(x) = z) = \sum_{y \in X} \mathbb{P}(f(x) = y) \cdot \mathbb{P}(f(y) = z).$$

If we rewrite the above with a measure-theoretic language, we see that we ought to define

$$f^2(x)(E) = \int_X f(y)(E)\ d(f(x))\{y\}.$$

More generally, given $f\colon X \to \mathcal{M}_1(Y)$, $g\colon Y \to \mathcal{M}_1(Z)$, we can define a "composition"

$$(g \circ f)(x)(E) = \int_X g(y)(E)\ d(f(x))\{y\}$$

so that $g \circ f\colon X \to \mathcal{M}_1(Z)$.

A map with type $X \to \mathcal{M}_1(Y)$ is called a *Markov kernel* between the measurable spaces X and Y. The category whose objects are measurable spaces and morphisms are Markov kernels, with compositions given by the rule above, will be called **Stoch**. We remark that the identity morphisms in this category are the Dirac masses $\delta(x) = \delta_x \in \mathcal{M}_1(X)$.

An interesting aspect of this category is that "deterministic" measurable maps still live inside it — given $g\colon X \to Y$ a measurable function, $f_g(x) = \delta_y$ is a Markov kernel. That is to say, this correspondence defines a (faithful) functor from the category **Meas** of measurable functions to **Stoch**. This means that dynamics in **Stoch** can be seen a generalization of deterministic measurable systems, where we now also allow the maps to be stochastic.

Another observation is that a probability measure μ in a measurable space X can be represented in this category by a Markov kernel $m\colon \{\star\} \to \mathcal{M}_1(X)$, where $\{\star\}$ is a measurable space with one element. Namely, we set $m(\star) = \mu$.

Iterated function systems

Invariant states

Given categories $\mathcal{C}, \mathcal{D}$, a *cone* of a functor $F\colon \mathcal{D} \to \mathcal{C}$ is defined as an object $C \in \mathcal{C}$ and a natural transformation $T\colon \Delta(C) \Rightarrow F$, where $\Delta(C)\colon \mathcal{D} \to \mathcal{C}$ is the constant functor at the identity of C. In the case we are interested, $\mathcal{D} = M$ is a monoid and F is given by the dynamics. This means a cone is given as an object $C \in \mathcal{C}$ and a morphism $c\colon C \to X$ with the property that for every $m \in M$,

$$\begin{tikzcd} & C \arrow[dl, "c"'] \arrow[dr, "c"] & \\ X \arrow[rr, "m"] & & X \end{tikzcd}$$

commutes.

A cone (C, c) is *universal* for $M \to \mathcal{C}$ if it is a limit of the above diagram, that is, for every other cone (D, d), there is a unique arrow $D \to C$ so that

$$\begin{tikzcd}
 & D \arrow[d, dashed] \arrow[dl, "d"'] \arrow[dr, "d"] & \\
 & C \arrow[dl] \arrow[dr] & \\
X \arrow[rr, "m"] & & X
\end{tikzcd}$$

commutes. Since categorical limits are unique, if a universal cone exists, it is unique up to isomorphism as well.

Example C.1.1. Let $f\colon X \to X$ be any function from a set to itself. Like in a previous example, this will define a dynamical system in **Set** over the monoid $\mathbb{Z}_{\geq 0}$, mapping $n \in \mathbb{Z}_{\geq 0}$ to the iterate f^n. If $x \in X$ is a fixed point of f, then we get a cone by defining $C = \{x\}$ and picking the morphism $c\colon C \to X$ to be the inclusion. Since x is a fixed point, it's clear that the associated diagram will commute. Conversely, if (C, c) is a cone and $\star \in C$, then the commutativity will mean $c(\star)$ is a fixed point for f.

The *universal* cone for f, then, will correspond to the set of all fixed points (which may be empty) paired with the set inclusion. Any other cone will map uniquely to this set, by matching each of its points with the corresponding fixed point for f.

Example C.1.2. Let $f\colon X \to \mathcal{M}_1(X)$ be a *Markov kernel* from a measurable set to itself. Then, examples of cones for the associated system in **Stoch** over $\mathbb{Z}_{\geq 0}$ are *stationary measures* $m\colon \{\star\} \to \mathcal{M}_1(X)$, that is, measures with the property that

$$(f \circ m)(\star) = \int_X f(x)\, d\mu\{x\} = \mu,$$

where $\mu = m(\star)$.

Extensions

The invertible extension

A failure of existence

C.2 Ergodic theorems

Below we reproduce a succinct proof of the pointwise ergodic theorem, described for example in [7, p. 32], [8, p. 136][1].

Theorem 24 (Birkhoff Ergodic Theorem)**.** *Let μ be an invariant probability measure and $\phi\colon X \to \mathbb{R}$ an integrable function. Then, for μ-almost every $x \in X$,*

$$\frac{1}{n}\sum_{i=0}^{n-1} \phi(f^i(x)) \to \mathbb{E}_\mu(\phi \mid \mathcal{I})(x),$$

where $\mathcal{I}$ is the sub-σ-algebra of (strictly) invariant sets.

Proof. Let

$$F_n(x) = \max_{1 \leq k \leq n} \sum_{i=0}^{k} \phi(f^i(x)).$$

[1]This is not exactly relevant to this dissertation, but the author has formally verified this proof in the Lean 4 ITP, and the code is currently available at `https://github.com/lucasvreis/BirkhoffErgodicThm`.

Claim 1: $F_{n+1} = \phi + \max\{0, F_n \circ f\}$.

To see this, note that $\sum_{i=0}^{k} \phi \circ f^i = \phi + \sum_{i=0}^{k-1} \phi \circ f^{i+1}$, so

$$F_{n+1} = \phi + \max_{1\le k\le n+1} \sum_{i=0}^{k-1} \phi \circ f^{i+1} = \phi + \max\Big\{0, \max_{1\le k\le n} \sum_{i=0}^{k} \phi \circ f^{i+1}\Big\} = \phi + \max\{0, F_n \circ f\}.$$

Claim 2: $F_{n+1} - F_n \circ f = \phi - \min\{0, F_n \circ f\}$.

This claim follows from the previous one by rewriting

$$F_{n+1} + \min\{0, F_n \circ f\} = \phi + \max\{0, F_n \circ f\} + \min\{0, F_n \circ f\} = \phi + F_n \circ f.$$

Two very useful corollaries of the claim are that for each $x \in X$, $F_{n+1}(x) - F_n(f(x))$ is a non-increasing sequence (because $F_n(f(x))$ is non-decreasing by definition), and that if $F_n(f(x)) \to \infty$, then the sequence will converge monotonically to $\phi(x)$.

Now, we define the set

$$S = \{x \in X \mid F_n(x) \to +\infty\}.$$

The first observation is that $S \in \mathcal{I}$, because it's measurable and strictly invariant. Since μ is invariant by f, this implies $\int_S F_n \circ f \, d\mu = \int_S F_n \, d\mu$. Also, the observations from the last claim show that $F_{n+1}(x) - F_n(f(x)) \to \phi(x)$ for $x \in S$. Thus, by the dominated convergence theorem,

$$0 \le \int_S F_{n+1} - F_n \, d\mu = \int_S F_{n+1} - F_n \circ f \, d\mu \to \int_S \phi \, d\mu \tag{C.1}$$

which means $\int_S \phi \, d\mu \ge 0$.

The second observation is that if $x \notin S$, then $(F_n(x))_n$ is bounded above, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \le \limsup_{n\to\infty} \frac{1}{n} F_n(x) \le 0. \tag{C.2}$$

Putting the two together, we conclude that if ϕ is such that $\mathbb{E}(\phi \mid \mathcal{I}) < 0$ almost everywhere, then by eq. (C.1) and the fact $S \in \mathcal{I}$, the set S has measure zero, so eq. (C.2) is valid almost everywhere.

The theorem then follows by noticing that for any $\varepsilon > 0$, if we replace ϕ by $\tilde{\phi} = \phi - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon$, then $\mathbb{E}(\tilde{\phi} \mid \mathcal{I}) < 0$, so

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \tilde{\phi}(f^i(x)) = \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) - \mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \le 0.$$

If we combine this with the analogous inequality for $-\phi$, we get

$$\mathbb{E}(\phi \mid \mathcal{I}) - \varepsilon \le \liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \le \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \phi(f^i(x)) \le \mathbb{E}(\phi \mid \mathcal{I}) + \varepsilon$$

and the result follows by letting $\varepsilon \to 0$. □

C.3 Empirical measures

Proposition 25. *Let X be a compact metric space with the Borel σ-algebra. For every $x \in X$, the limit points in the weak* topology of the sequence of empirical measures*

$$\mathcal{M}(x) = \{e_n(x)\}'_{n\geq 0} = \bigcap_{i=0}^{\infty} \overline{(\{e_n(x)\}_{n\geq i})}$$

form a compact connected subset of the invariant measures.

Proof. The set is closed by definition, so by Banach-Alaoglu it's compact; for connectedness, suppose $\mathcal{M}(x) = K_1 \cup K_2$ where K_1, K_2 are closed and disjoint. Let $\{\phi_i\}_{i\in\mathbb{N}} \subseteq B_{C(X)}$ be a dense subset and define

$$d(\mu, \nu) = \sum_{i=1}^{\infty} \frac{|\mu(\phi_i) - \nu(\phi_i)|}{2^i},$$

which is a metric inducing the weak* topology on the set of probability measures. Since it's a metric, there are two open sets $U_1 \supseteq K_1, U_2 \supseteq K_2$ with positive distance $s > 0$ from each other. We claim $\lim_{n\to\infty} d(e_n(x), e_{n+1}(x)) = 0$, which would contradict $s > 0$ because the definition of $\mathcal{M}(x)$ implies $e_n(x) \in U_1$ and $e_{n+1}(x) \in U_2$ for infinitely many n. Indeed,

$$\begin{aligned} d(e_n(x), e_{n+1}(x)) &= \sum_{i=1}^{\infty} \frac{|\frac{1}{n}\sum_{i=0}^{n-1} \phi_i \circ f^i(x) - \frac{1}{n+1}\sum_{i=0}^{n} \phi_i \circ f^i(x)|}{2^i} \\ &\leq \sum_{i=1}^{\infty} \frac{2}{n}\frac{\|\phi_i\|_\infty}{2^i} \leq \frac{2}{n} \to 0. \end{aligned}$$

□

C.4 Skew products

Objects

W 13

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